

Smart Inventory & Warehouse Automation Platform

This document describes each of the six workflows that make up the platform in detail: what it is for, how it is triggered, which sheets and external services it touches, which workflows it calls, and a walkthrough of every node it contains.

WF1: Inventory Monitoring & Stock Updates

Continuously watches stock levels across all products and kicks off the reorder process the moment something falls below its reorder point, without anyone having to check manually.

Trigger: Cron, every 30 minutes

Sheets: Reads Inventory_Master

Inter-workflow calls: Calls WF2 (low-stock-trigger) when low stock is found, calls WF6 (audit-log) after every run

Node count: 6

Node-by-node walkthrough

#	Node	Type	Function
1	Cron Trigger	Cron	Fires the workflow every 30 minutes.
2	Read Inventory_Master	Google Sheets	Reads every row of the Inventory_Master sheet.
3	Check Low Stock	Code	Filters items where current_stock is less than or equal to reorder_point.
4	Any low stock?	If	Branches on whether the filtered list has one or more items.
5	Trigger WF2	HTTP Request	Posts the low stock items to WF2's webhook if the list is non-empty.
6	Log Scan	HTTP Request	Posts a SCAN summary (items scanned, low stock count) to WF6 regardless of outcome.

WF2: Low Stock Detection & Purchase Request

Turns a list of low stock items into properly formed, AI reasoned purchase orders, ready to be routed for approval.

Trigger: Webhook: /low-stock-trigger

Sheets: Reads Supplier_Registry, writes to Purchase_Orders

Inter-workflow calls: Calls WF3 (po-approval) once each PO is recorded

Node count: 8

Node-by-node walkthrough

#	Node	Type	Function
1	Webhook Trigger	Webhook	Receives the low stock item list posted by WF1.
2	Item Lists	Item Lists	Splits the incoming array into one item per execution branch.
3	Get Supplier Info	Google Sheets	Looks up the relevant supplier's details, including average delivery time.
4	Groq Chat Model	LangChain LLM (Groq)	Provides the underlying language model (llama-3.3-70b-versatile) used by the chain below.
5	Basic LLM Chain	LangChain Chain	Prompts the model with stock levels and supplier lead time, asking for a recommended order quantity and reasoning as JSON.
6	Create PO Record	Code	Parses the AI's JSON response, falls back to a max-stock-minus-current-stock calculation if parsing fails, and builds the PO object.
7	Append to Purchase_Orders	Google Sheets	Appends the new PO record as a row in Purchase_Orders.
8	Trigger WF3	HTTP Request	Posts the PO to WF3's webhook for approval routing.

WF3: Supplier Communication & Approval

Adds a human control point on spend: high value orders require a manager's sign-off before a supplier is contacted, while smaller orders proceed automatically.

Trigger: Webhook: /po-approval

Sheets: Updates Purchase_Orders

Inter-workflow calls: None outbound to other workflows; sends Telegram messages and a Gmail email

Node count: 8

Node-by-node walkthrough

#	Node	Type	Function
1	Webhook Trigger	Webhook	Receives the PO record posted by WF2.
2	IF > \$5000	If	Branches on whether estimated_cost exceeds the 5000 dollar threshold.
3	Telegram Request Approval	Telegram	Sends an approve/reject message with resumeUrl links for orders above the threshold.

#	Node	Type	Function
4	Wait	Wait (webhook resume)	Pauses execution until the manager clicks one of the links in the Telegram message.
5	Is Approved?	If	Reads the action query parameter from the resumed webhook to branch on approve vs reject.
6	Update PO Status (Approved)	Google Sheets	Sets the PO's status to Approved in the Purchase_Orders sheet.
7	Update PO Status (Rejected)	Google Sheets	Sets the PO's status to Rejected in the Purchase_Orders sheet.
8	Send PO Email	Gmail	Emails the supplier with the order quantity, product, and estimated cost once approved.

WF4: Delivery Tracking & Inventory Update

Reconciles an incoming delivery against its original purchase order and brings the inventory sheet up to date, closing the loop from purchase to stock.

Trigger: Webhook: /delivery-received

Sheets: Reads Purchase_Orders and Inventory_Master, writes to Inventory_Master and Delivery_Log

Inter-workflow calls: None; notifies over Telegram

Node count: 7

Node-by-node walkthrough

#	Node	Type	Function
1	Webhook Trigger	Webhook	Receives the delivery event (po_id, quantity_delivered) from the external delivery system.
2	Lookup PO	Google Sheets	Finds the matching purchase order by po_id.
3	Qty Matches?	If	Compares quantity_delivered against the PO's order_quantity.
4	Get Current Stock	Google Sheets	Fetches the product's current stock row from Inventory_Master.
5	Update Inventory	Google Sheets	Adds the delivered quantity to current_stock and writes it back.
6	Append to Deliveries	Google Sheets	Logs the delivery as a new row in Delivery_Log.
7	Telegram Notification	Telegram	Sends a confirmation message with the PO id, product, and quantity received.

WF5: Inventory Analytics & Reports

Gives leadership a recurring, AI written snapshot of inventory health without anyone needing to compile it manually.

Trigger: Cron, weekly

Sheets: Reads Inventory_Master

Inter-workflow calls: None; sends a Gmail report

Node count: 6

Node-by-node walkthrough

#	Node	Type	Function
1	Cron Trigger (Weekly)	Cron	Fires the workflow once a week.
2	Read Inventory	Google Sheets	Reads the full Inventory_Master sheet.
3	Aggregate Metrics	Code	Computes counts of low stock items and overstocked items (over 120 percent of max level).
4	AI Generate Insights	LangChain LLM (Groq)	Provides the model (llama3-70b-8192) used by the chain below.
5	Basic LLM Chain	LangChain Chain	Prompts the model to write a brief, professional report analyzing the stock summary and raw data.
6	Email Report	Gmail	Emails the generated report to the inventory analyst distribution.

WF6: Audit Trail & Error Handling

Acts as the platform's shared logging and error handling hub, giving every workflow a single place to record what happened and to raise a critical alert when something breaks.

Trigger: Webhook: /audit-log, plus a built-in Error Trigger

Sheets: Writes to Audit_Trail

Inter-workflow calls: None; sends a Telegram alert for critical errors

Node count: 7

Node-by-node walkthrough

#	Node	Type	Function
1	Webhook Trigger	Webhook	Receives explicit log events posted by other workflows, such as WF1's scan summaries.
2	Format Log Entry	Code	Builds a structured audit record with an event id, timestamp, workflow, event type, and severity.
3	Append to Audit_Trail	Google Sheets	Appends the formatted entry to the Audit_Trail sheet.
4	Error Trigger	Error Trigger	Fires automatically when a workflow configured to use WF6 as its error workflow fails.
5	Format Error Details	Code	Builds a structured audit record from the error payload, tagged with CRITICAL severity.
6	Append Error to Audit_Trail	Google Sheets	Appends the formatted error entry to the Audit_Trail sheet.
7	Telegram Alert	Telegram	Sends an immediate alert message naming the failing workflow and the error detail.